CCA

First introduced by Hotelling in 1936. Suppose we have two sets of data $\mathbf{X} \in \mathbb{R}^{n \times p}, \mathbf{Y} \in \mathbb{R}^{n \times q}$ where n represents the number of observations, so we can understand these as n i.i.d. realizations of the mean-zero random vectors $X \in \mathbb{R}^p$ and $Y \in \mathbb{R}^q$. Our goal is that of finding directions $\mathbf{a} \in \mathbb{R}^p$, $\mathbf{b} \in \mathbb{R}^q$ such that the correlation of $\mathbf{a}^\top X$ and $\mathbf{b}^\top Y$ is maximized.
Let $\Sigma_X = \mathrm{Var}(X)$, $\Sigma_Y = \mathrm{Var}(Y)$ and $\Sigma_{XY} = \mathrm{Cov}(X, Y) = \mathbb{E}[XY^\top]$, then we wish to maximize the expression

$$\rho(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a}^\top \Sigma_{XY} \mathbf{b}}{\sqrt{\mathbf{a}^\top \Sigma_X \mathbf{a} \mathbf{b}^\top \Sigma_Y \mathbf{b}}}.$$

To solve such a problem, we take the change of variables $\alpha = \Sigma_X^{1/2} \mathbf{a}$ and $\beta = \Sigma_Y^{1/2} \mathbf{b}$, then the correlation becomes

$$\rho(\alpha, \beta) = \frac{\alpha^\top \Sigma_X^{-1/2} \Sigma_{XY} \Sigma_Y^{-1/2} \beta}{\|\alpha\| \|\beta\|} = \frac{\alpha^\top M \beta}{\|\alpha\| \|\beta\|} = \left(\frac{\alpha}{\|\alpha\|}\right)^\top M \left(\frac{\beta}{\|\beta\|}\right),$$

for $M = \Sigma_X^{-1/2} \Sigma_{XY} \Sigma_Y^{-1/2} \in \mathbb{R}^{p \times q}$. Therefore, our resulting optimization is the linear algebra problem

$$\max_{\|\alpha\| = \|\beta\| = 1} \alpha^\top M \beta.$$

We can now find the SVD of the matrix $M = UDV^\top$ where U, V are (semi)orthogonal matrices to obtain

$$\max_{\|\alpha\| = \|\beta\| = 1} \left(U^\top \alpha\right)^\top D \left(V^\top \beta\right) = d_1(M),$$

where the arguments that produce this for α and β are the first left and right singular vectors respectively. The corresponding pair $\mathbf{a}_1, \mathbf{b}_1$ are termed *canonical correlation vectors* or *canonical directions*. In fact, for each singular values d_i we can find the corresponding vectors $\mathbf{a}_i, \mathbf{b}_i$ to the ith canonical correlation vectors α_i, β_i. In this way we obtain the vectors

$$\mathbf{a}_1, \mathbf{a}_2, \ldots, \mathbf{a}_r, \quad \mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_r,$$

for r the rank of M, which solve the constrained optimization problem

$$\max_{\mathbf{a}, \mathbf{b}} \mathbf{a}^\top \Sigma_{XY} \mathbf{b}, \quad \text{s.t.} \quad \mathbf{a}^\top \Sigma_X \mathbf{a} = \mathbf{b}^\top \Sigma_Y \mathbf{b} = 1,\, \mathbf{a}^\top \Sigma_X \mathbf{a}_i = \mathbf{b}^\top \Sigma_Y \mathbf{b}_i,\, \forall i < j.$$

Related discussion in ESLII 3.7